



Insertion Sort

AN UNPLUGGED ACTIVITY

Unplugged Activity:

Sorting some playing cards

1. Start with an empty left hand and 5 cards are facing down on the table.
2. Remove **one card** from the table, and place it at the last card in the left hand.
3. Compare this card with its adjacent card; swap if necessary until all the cards in the hand are sorted.
4. Repeat step 2 and 3 until you get 5 sorted cards held in the left hand.

Unplugged Activity:



**We have 5 cards of
unknown values.**



First card : 10

Second card : 6



Second card : 6

Swap to place them in order



Third card : 36

Since 36 is larger than 10,
no swapping is required.



Forth card : 24

Swap to place them in order

Check whether further
swapping is required



Fifth card : 12



Fifth card : 12

Make multiple swap until they are placed in order



Fifth card : 12

Make multiple swap until they are placed in order



Done !

We have a sorted set of cards when there is no more swapping.



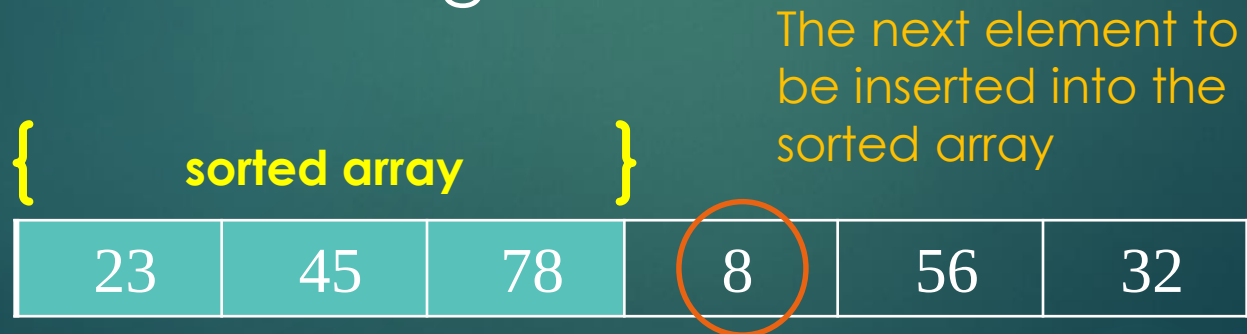
Insertion Sort

ALGORITHM

Description :

Insertion Sort is a simple sorting algorithm that works the way we sort playing cards in our hands. We sort the first two cards and then place the third card in the appropriate position within the first two, and then the fourth is positioned within the first three, and so on until the whole hand is sorted.

During an iteration, an element of the list is inserted into the sorted portion of the array to its left. So, basically, for each iteration, we have **an array of sorted elements** to the left, and an array of other elements still to be sorted to the right.



Sorted

Unsorted



23	78	45	8	32	56
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Original List



23	78	45	8	32	56
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After pass 1



23	45	78	8	32	56
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After pass 2



8	23	45	78	32	56
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After pass 3



8	23	32	45	78	56
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After pass 4

8	23	32	45	56	78
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After pass 5

Basic skill #1:

For a list containing only 2 elements, swap the elements, if necessary, such that the smaller element will be shifted to the left.

Basic skill #2:

For a list with all the elements sorted in ascending order except the last element, comparing two elements at a time, and swapping when necessary, arrange the elements in an ascending order.

Insertion Sort:

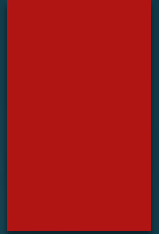
For an unsorted list, perform Basic Skill #1 on a first sub-list of 2 sorted elements, then use the Basic Skill #2 to get a sub-list of 3 sorted elements, repeat using Basic Skill #2 ... until the whole list is sorted in an ascending order.

In-Place Insertion Sort



This algorithm is efficient for smaller lists, but very inefficient for larger lists. Also, it does not require any additional storage space, as it operates in-place.

Order of Growth



Insertion Sort : $O(n^2)$

For an array of n unsorted items, to shift one element to the 'correct' position, the Insertion Sort algorithm needs to scan through the whole array of $(n-1)$ terms, hence the Order of Growth is $O(n^2)$.